



The Data Foundation Sheet

Getting the question right, reading data in reproducibly, separating defects from evidence, and building the features a model needs. Work down the numbered stages. Each answers one question.

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1

FRAME What decision changes because of this work, and who makes it?

Fix these four first

- ▶ **Unit of analysis.** One row equals what: a request, a customer, a customer-month.
- ▶ **Target definition.** Plain words, plus the exact SQL that produces it.
- ▶ **Decision point.** When the prediction is used. Nothing recorded after it can be a feature.
- ▶ **Cost of each error.** False positive against false negative. This picks the metric at Stage 7.

Identify the problem shape

- ▶ **Describe** what happened: SQL, marts, dashboards. No model needed.
- ▶ **Diagnose** why: exploratory analysis, statistical tests, cohorts.
- ▶ **Predict** what next: supervised learning. Needs labelled history.
- ▶ **Prescribe** what to do: causal inference, uplift, optimisation. Correlation is insufficient.

NOTE

Check before proceeding

- ▶ If a lookup reproduces the target exactly, it is a business rule and there is nothing to learn.
- ▶ Name the person who will act on the output. Record it with the target definition.
- ▶ Where a query answers the question, use the query.

2

LOAD Where does it live, how large is it, and how is the read made repeatable?

SOURCE	USE	REASON
Delimited text	<code>read_csv</code> with explicit <code>dtype=</code> , <code>parse_dates=</code> , <code>na_values=</code>	Inference varies with file contents. Declared types make the read deterministic.
Spreadsheet	<code>read_excel</code> with <code>openpyxl</code> , one call per named sheet	Merged cells, hidden rows and display formatting are not data. Validate the row count.
REST API or JSON	<code>requests</code> , then <code>json_normalize</code>	Nested payloads need flattening to a stable schema. Cache the raw response to repeat offline.
Relational database	SQLAlchemy engine, or a dbt source	Filter and join in the database. Select only the columns used.
Columnar (Parquet)	<code>pyarrow</code>	Types and compression travel with the data. Default for anything passed between steps.
Larger than memory	DuckDB or Polars. Dask or Spark only if genuinely distributed	Single-node columnar engines handle tens of gigabytes without cluster failure modes.
Streaming or event	Land to object storage in batches, then treat as files	Micro-batch keeps the analytics path testable with the same tools.

CHECK YOUR OUTPUT

Confirm before Stage 3

- ✓ Row count matches source, recorded as a number, not glanced at with `.head()`.
- ✓ `df.dtypes` printed and compared against the declared schema.
- ✓ Raw layer unchanged. Every later step writes a new table.
- ✓ Encoding set explicitly to `utf-8`.
- ✓ Exact command recorded so the step reruns from a log.
- ✓ Expected row count per source captured on the first clean run. Investigate any later difference before proceeding.

3

CLEAN Which values are defects, which are artefacts, and which must be kept?

Missingness: classify, then treat

- ▶ **Completely at random.** No pattern. Drop if few, impute with median or mode if many.
- ▶ **At random.** Explained by other columns. Impute by model or grouped statistic.
- ▶ **Not at random.** Absence carries information. Add an `is_missing` flag and keep it as a feature.
- ▶ Fit imputers inside the fold, never on the full dataset.

Duplicates: three distinct cases

- ▶ **Exact rows.** Safe to drop. Record the count dropped.
- ▶ **Same key, differing detail.** Apply a stated rule: latest wins, or most complete wins. Document it.
- ▶ **Same event, different key.** Needs fuzzy matching. Full-row deduplication misses these.
- ▶ Count duplicates by business key as well as by row. The two counts rarely agree.

NOTE

Outliers are not automatically errors

- ▶ **Impossible** (negative age, close before open): correct or remove.
- ▶ **Implausible** (resolved in zero hours yet counted as meeting the target): investigate and record the finding.
- ▶ **Extreme but valid** (one genuine major incident): keep. Use a robust model or metric.
- ▶ Do not remove records because they reduce the score.

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ENGINEER FEATURES What does the model need that the raw columns do not already express?

NOTE

Leakage: stop list

- ▶ Any column recorded after the decision point, including closure fields and outcome timestamps.
- ▶ Scaler, imputer, encoder or selector fitted before the split.
- ▶ Oversampling before the split, putting copies of a row on both sides.
- ▶ Target encoding computed across the full dataset.
- ▶ An identifier that happens to sort by outcome.
- ▶ Duplicate records split across train and test.
- ▶ **Test:** a score well above expert expectation is leakage until shown otherwise.

Feature recipes

- ▶ **Time deltas.** Days since previous event, age at the decision point.
- ▶ **Calendar context.** Working day, holiday, month end, position in quarter.
- ▶ **Rolling aggregates.** Mean, count, maximum over a trailing window, past rows only.
- ▶ **Ratios.** Per head, per unit, share of parent total.
- ▶ **Interactions.** Category crossed with an amount band, where the business treats them together.
- ▶ **Text flags.** Length, keyword presence, proportion of capitals.
- ▶ **Rarity.** Frequency of the level in training data.
- ▶ **Deviation.** Value minus group mean.

Define features in SQL, not in a notebook. A feature defined in a transformation model is versioned, tested, documented and available to every consumer. A feature defined in a cell exists until the kernel restarts.

The Data Type Reference Sheet

What kind of quantity a column holds determines what may be done to it: how to clean it, how to encode it, whether to scale it, how to chart it, and which test guards it. This sheet is the lookup for that decision.

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5 MAP THE TYPES What kind of quantity is each column, and what operations does that permit?

LOGICAL TYPE	STORE AS	CLEAN: WATCH FOR	PREPARE FOR MODELLING	SCALE	PRESENT AS	GUARD WITH
Nominal category no order	category	Case and whitespace variants of one level; levels unseen at scoring time; high cardinality	One-hot below about 15 levels. Target or frequency encoding above that, fitted in the fold. Native categorical in LightGBM or CatBoost.	No	Bar, count plot	accepted_values, not_null
Ordinal category ranked	ordered categorical	Order lost on read; unequal gaps treated as equal; a grade inserted mid-scale	Integer map preserving rank. One-hot discards the order.	Optional	Ordered bar	accepted_values
Binary flag	bool, int8	Mixed encodings (Y, yes, 1, true); nulls cast to True; three states reduced to two	Map to 0 and 1. Keep unknown as its own level where it carries meaning.	No	Stacked bar	accepted_values
Count	Int64 nullable	Negatives; zero conflated with missing; counts over different exposure windows compared directly	log1p if skewed. Normalise by exposure. Poisson objective for a count target.	Linear only	Histogram	Expression >= 0
Continuous ratio true zero	float64	Impossible zeros and negatives; mixed units; sentinels such as 999 or minus 1; long right tail	Log, Box-Cox or Yeo-Johnson for skew. Winsorise rather than delete. Bin only where the relationship is non-monotonic.	Yes	Histogram, box	not_null, range
Interval no true zero	float64	Ratios computed on the values (20 degrees is not twice 10); scales mixed across sources	Differences, not ratios. Convert to one scale in staging.	Yes	Line	Range
Timestamp	datetime64[ns, tz]	Naive and aware mixed; DST gaps; epoch seconds read as milliseconds; dates in the future; day-first against month-first	Derive hour, day of week, month, quarter, is-holiday, days-since. Cyclical parts as sine and cosine.	Parts only	Time series line	not_null, freshness
Duration	timedelta64, then hours	Negative durations where close precedes open; zero-length records treated as successes; unit ambiguity	Convert to one unit in staging and name the column for that unit.	Yes	Histogram	Expression > 0
Identifier	string	Read as integer and used as a feature; leading zeros stripped; reused across systems	Never a feature. Joining, grouping and splitting only.	Never	Not plotted	unique, not_null
Free text	string	Empty string and null treated differently; personal data; encoding damage; boilerplate dominating content	Length, keyword and pattern flags first. Then TF-IDF. Then embeddings if the cost is justified.	n/a	Term frequency	not_null on the flag
Monetary	float64; Decimal for ledgers	Mixed currencies in one column; thousands separators read as text; floating point rounding in reconciliation	One currency at a dated rate. Deflate for inflation. Express per head or per unit.	Yes	Bar	Expression >= 0
Proportion or rate	float64	Zero-to-one and zero-to-100 mixed in one column; rates built on very small denominators	Logit transform for modelling. Carry the denominator as its own column.	Yes	Histogram	Range 0 to 1
Geospatial	float64 pair, or code	Latitude and longitude transposed; truncated postcodes; boundaries changed between periods	Distance to a point of interest, spatial join to an area, then encode the area as a category.	Yes	Map, choropleth	Range
Hierarchy or code	category	Codes reissued over time; ragged depth; rare levels holding two or three rows	Model at the level the decision is taken. Group rare levels into a named Other.	No	Treemap, bar	relationships

Scale means centre and scale before fitting. Required for distance-based and gradient-based methods: k-nearest neighbours, support vector machines, k-means, principal component analysis, penalised linear and logistic regression, neural networks. Not required for decision trees, random forests or gradient boosting. Guard with names the transformation test that should fail the build if the rule is broken.



The Method Selection Sheet

Whether the problem is supervised, unsupervised or neither yet, and which family of algorithm to reach for once that is settled. Start at the root question and follow the branch that matches the data.

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SELECT THE LEARNING TYPE Is the outcome recorded for enough past cases, as at the decision point?

START HERE

Is there a recorded outcome that was not knowable at the decision point, available for a usable share of rows?

YES, LABELLED SUPERVISED

Continuous target. Regression: hours, cost, demand.

Two classes. Binary classification: will this breach?

Many exclusive classes. Multiclass: which team should own this?

Several labels at once. Multilabel: independent tags per row.

Ranked classes. Ordinal regression: preserves the order.

Counts. Poisson or negative binomial regression.

Time to event, some cases open. Survival analysis. Classification discards censored rows and biases the estimate.

Ordered in time. Forecasting: classical methods, or boosted trees on lag features.

NO LABELS UNSUPERVISED

Group similar rows. Clustering. k-means for round, similarly sized groups. HDBSCAN for irregular shapes, noise and unknown k. Gaussian mixtures for soft membership. Hierarchical where a dendrogram is needed.

Too many columns. PCA for linear compression. UMAP for structure. t-SNE for pictures only, never as model input.

Identify unusual records. Isolation Forest, Local Outlier Factor, One-Class SVM. Also applies where positives fall below one per cent.

What occurs with what. Association rules: basket and co-occurrence.

Themes in text. Topic modelling, or clustering over embeddings.

PARTLY, OR A DIFFERENT QUESTION NEITHER, YET

Few labels, many rows. Semi-supervised or self-training. Label the most informative rows first through active learning.

Labels are rules or noisy. Weak supervision: combine several imperfect labelling functions.

The effect of intervening is needed. Causal inference or uplift. A predictive model finds who will breach, not who can be helped.

A sequence of decisions with feedback. Reinforcement learning. Needs a simulator or a safe evaluation route first.

Rich input, small labelled set. Transfer learning: pretrained encoder, small head trained on top.

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SELECT THE ALGORITHM What shape is the data, and what must be explained to those acting on the output?

SITUATION	START WITH	REASON	TRADE-OFF
Every project, before anything else	A constant predictor, then linear or logistic regression	Lift can only be stated against a published baseline. A well-specified linear model is often competitive.	Adds a short step at the start of every project.
Tabular, mixed types, thousands to millions of rows	Gradient boosting: LightGBM, XGBoost, CatBoost	Handles interactions, missing values and unscaled inputs without extra preparation.	Needs care with categorical encoding and early stopping.
Fewer than about a thousand rows	Penalised linear or logistic regression, naive Bayes, k-nearest neighbours	High-capacity models fit noise in small samples. Regularisation controls this directly.	Repeated cross-validation required. A single split is uninformative.
Must be explained or audited	Logistic regression on binned inputs, shallow tree, generalised additive model, explainable boosting	Sign and magnitude of each effect can be read and presented for challenge.	Usually a small reduction in predictive performance.
Wide sparse text features	Linear SVM or logistic regression on TF-IDF	Strong and fast on document classification. Tree methods struggle in very high dimensions.	No word order or sense. Move to embeddings if that matters.
Images, audio, long documents	A pretrained model, fine-tuned	Transfer learning outperforms training from scratch at typical data volumes.	Hardware, longer cycles, reduced explainability.
Strong seasonality and trend	Classical time series, or boosted trees on lag and calendar features	Row-independent models cannot see temporal structure unless it is supplied as features.	Trees do not extrapolate a trend. Detrend first.
Rare positives, below about five per cent	Boosting with class weights and a tuned threshold, or reframe as anomaly detection	Weighting and threshold selection address the operational decision directly.	Synthetic oversampling is a last resort, training fold only.
Several comparable models, no clear leader	Average or stack them	Averaging models with uncorrelated errors reduces variance.	Harder to explain, deploy and debug.

COMMON ERRORS

CHECK FOR THESE

- ▶ A score well above expert expectation. Treat as leakage until shown otherwise.
- ▶ The test set consulted more than once. It is now a validation set and the final figure is optimistic.
- ▶ A figure in a report that cannot be regenerated from a recorded command.
- ▶ Accuracy quoted on an imbalanced problem.
- ▶ Preprocessing fitted before the split.
- ▶ Feature importances presented as causes. They are associations within one fitted model.
- ▶ Records removed because they reduce the score.
- ▶ A model in production with no owner, no monitoring, no retraining trigger.
- ▶ A notebook used as the production artefact.
- ▶ Tuning continued after the gain falls inside the fold-to-fold spread.



The Evaluation and Delivery Sheet

Splitting and scoring honestly, then moving a result into a tested, scheduled pipeline that still holds six months later. Terms and abbreviations are defined at the foot of this sheet.

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VALIDATE AND SCORE Does the result hold on data the model has not seen?

DATA STRUCTURE	SPLIT THIS WAY	WHAT MATTERS	SCORE WITH
Independent rows, balanced classes	Stratified k-fold, five or ten folds	Regression, error in real units	MAE . Robust, directly interpretable
Several rows per entity	Grouped k-fold on the entity key, so no entity spans the split	Regression, large errors costly	RMSE . Penalises large errors more heavily
Time matters, or future rows will be scored	Forward chaining: train on past, test on future, with a gap where labels arrive late	Regression, relative error	MAPE . Invalid near zero; use a scaled alternative
Very small sample	Repeated stratified cross-validation, or leave one out	Ranking cases for a work queue	ROC AUC . Threshold independent; measures separation
Tuning plus a final unbiased estimate	Nested cross-validation, or a held-out test set used once	Rare positives	PR AUC and recall at a fixed workload. ROC AUC is optimistic here
Deployment across regions or sites	Leave one group out, to test generalisation to an unseen site	A false alarm is costly	Precision at the operating threshold
		A miss is costly	Recall , reported with the precision it costs
		The probability itself is used	Log loss or Brier score , with a calibration curve
		Many classes	Macro F1 where classes matter equally, weighted where prevalence does. Read the confusion matrix either way
		An operational decision	Expected value from a cost matrix, in the units the business uses
		Clusters	Silhouette , with a domain review of the groups

Accuracy applies only where classes are balanced and both error types cost the same. Quote every metric with the baseline it beats and the spread across folds. A single figure without both is not a result.

CHECK YOUR OUTPUT
Reproducibility requirements

- ✓ Pin every version in requirements.txt.
- ✓ Set the seed, and sort the dataframe into a canonical order before splitting. A database returns rows in physical order, which can change between builds, so a fixed seed alone does not guarantee the same split.
- ✓ Put preprocessing inside a pipeline object, so it fits per fold and deploys with the model.
- ✓ Report the spread across folds alongside the mean.

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OPERATIONALISE Will this run, stay correct, and stay trusted in six months?

Layer the transformation

- ▶ **Sources**. Raw, unmodified, freshness tested.
- ▶ **Staging** `stg_`: one model per source table. Rename and cast only. No joins, no logic.
- ▶ **Intermediate** `int_`: joins and business logic, each step separately testable.
- ▶ **Marts** `fact_` and `dim_`: the models consumers query.
- ▶ **Features**: a versioned model, so training and scoring read one definition.

Test and orchestrate

- ▶ Add a test for every defect found. Generic first: `unique`, `not_null`, `accepted_values`, `relationships`. Then expressions for business rules.
- ▶ Record pass, warn and error counts from every build. Treat a new warning as a change needing explanation.
- ▶ **Airflow** where the unit of work is a scheduled task graph, with mature operators and alerting.
- ▶ **Dagster** where the unit of work is the asset produced and lineage is the primary concern.
- ▶ Run steps as subprocesses where dependencies conflict. This also records a reproducible command per task.

NOTE

After deployment

- ▶ **Input drift**. Compare live feature distributions against training. PSI above 0.25 needs investigation.
- ▶ **Prediction drift**. Monitor the score distribution. It moves before measured performance does.
- ▶ **Performance**. Measurable only once labels arrive. Establish that lag before promising a reporting schedule.
- ▶ **Slices**. Report performance by group as well as overall.
- ▶ **Feedback**. Where acting changes the outcome, future labels are affected. Keep a control group.

K

KEY: TERMS AND ABBREVIATIONS

AUC Area under a curve; one number for a curve of trade-offs.	Mean absolute percentage error; invalid near zero.	Lineage Traced path from source column to output.	image or an item.	Non-linear projections; cluster distances are not meaningful.
ROC True positive rate against false positive rate, all thresholds.	R2 Share of variance explained; not correctness.	Mart A curated, business-facing model.	Winsorise Cap extremes at a percentile instead of deleting.	HDBSCAN Density clustering; irregular shapes, labels noise.
PR Precision against recall; preferred when positives are rare.	CV Cross-validation; repeated splits to estimate spread.	Grain What exactly one row represents.	MCAR, MAR, MNAR Missing completely at random; at random; not at random.	SMOTE Synthetic minority oversampling; training fold only.
Precision Of cases flagged, the share correct.	Leakage Training information unavailable at prediction time.	Fact, dimension Measured events, and the context they are measured against.	IID Independent and identically distributed.	Class weight Makes errors on one class count for more.
Recall Of cases that mattered, the share found.	Overfitting Fitting the sample rather than the pattern.	SCD Slowly changing dimension; keeps history as attributes change.	Censoring The event has not yet happened for that row.	Threshold Probability cut-off turning a score into an action.
F1 Harmonic mean of precision and recall.	Regularisation A penalty on complexity; trades bias for variance.	OHE One-hot encoding; one binary column per level.	GBM Trees built in sequence, each correcting the last.	SHAP Per-prediction feature contribution.
Brier Mean squared error of predicted probabilities.	Baseline The trivial model a result must beat.	Target encoding A level replaced by its mean outcome; fit in the fold.	GLM Linear model fitted for non-normal outcomes.	Uplift Change in outcome caused by acting.
Calibration Whether a stated 70 per cent happens 70 per cent of the time.	ELT Extract, load, transform; the transform runs in the warehouse.	TF-IDF Weights words by how distinctive they are.	GAM A smooth curve per feature; still readable.	PSI Population stability index; a drift measure.
MAE Mean absolute error; in target units.	DAG Directed acyclic graph; dependencies with no cycles.	Embedding A dense vector representing text, an	SVM Support vector machine.	Drift Live data no longer resembling training data.
RMSE Root mean squared error; weights large errors.	Asset A persistent object a pipeline produces, such as a table.		kNN k-nearest neighbours.	SLA The standard a case is measured against.
MAPE			PCA Linear dimensionality reduction.	
			UMAP, t-SNE	